HITACHI

Hitachi Rail STS

V300 ZEFIRO

A New Sense of Speed

Train Protection-VF.
CONCEPT
SFRS

Responsible Division:	Responsible Unit:	Document Type:	Distribution Status:	Document State:

	Name	Signature	Date
Prepared:	LUCA GALGANO Prepared		07/07/2022
Checked:	DANIELE FERRARI System Engineering Coordinator		07/07/2022
Approved:	DANIELE FERRARI Head of Vehicle Engineering		07/07/2022
Released:	GIUSEPPE CARRINO Deputy Project Engineer		07/07/2022

HITACHI

Hitachi Rail STS

Identity Number:

FA020013231

Effective Date:	Revision:	Language:
07/07/2022	01	En

Revision log

Revision	Date	Changes	Reason
00	22/02/2022	First issue	
01	07/07/2022	Second release	• §3.1.2.13: Added requirements for switchable data entry function • Modified 2F_09.02.Zefiro-Europe.CONCEPT.672 (error in the number of the reference requirement)

Content

1 Introduction 4

1.1 Train Configuration 4

1.2 Definitions and abbreviations 4

2 System Overview 5

2.1 System Configuration 5

3 Functional requirements 6

3.1 Train Protection 6
3.1.1 Driver´s Safety Device (DSD) 6
3.1.2 Automatic Train Protection System (ATP) 10
3.1.3 Juridical Recorder Unit (JRU) 24
3.1.4 Cab Radio 26

4 Signal Interface 27

4.1 I/O Interface 27
4.1.1 Supplier system 27
4.1.2 TCMS SW 28

4.2 Bus Interface 29

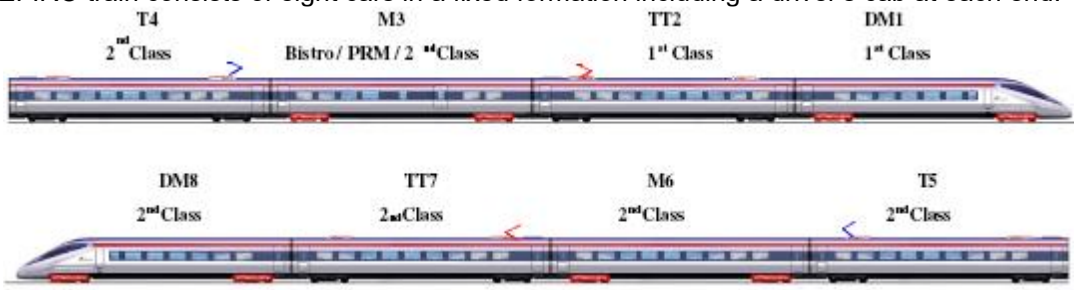
5 References 29

1 Introduction

This document is a function specification for the control of the *ATP* system for new V300ZEFIRO Italy (equipped with ETCS BL3) for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

Acronym	Meaning
CD	Circuit Diagram
CEA	Commutatore Esclusione ATP (ATP Exclusion Switch)
CETD	Commutatore Escusione Tagio trazione DIS (DIS traction cut off exclusion switch)
DIS	Driver Information system
DVD	Driver Vigilance Device
EB	Emergency Brake
ETCS	European Train Control System
EVC	European Vital computer
EVENT	Diagnostics that will be stored in the Event database within TCMS SW
EVIG	Esclusione Vigilante (DVD exclusion switch)
FS	Full supervision
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps
IDU	Interface Display Unit
JRU	Juridical Recorder Unit
MMI	Man Machine Interface
NSTOP	Non stopping area
RSC	Ripetizione Segnali Continua (Italian National ATP)
SCMT	Sistema Controllo Marcia Treno (Italian National ATP)
SR	Staff Responsible
TCMS	Train Control and Management System
TN	Tratto Neutro (Neutral Section)
TT	Taglio Trazione (Traction Cut off)

2 System Overview

2.1 System Configuration

The following figure shows the architecture of the ATP system:

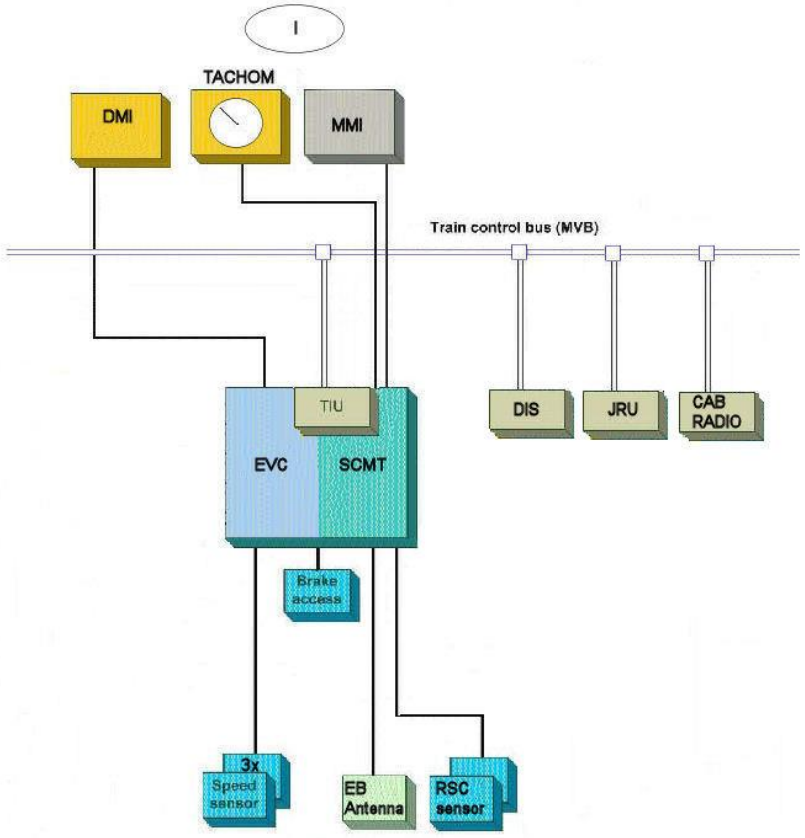


Figure 1: Italy configuration

ATP equipment:

- ETCS BL3 (SCMT integrated)
- Cab Radio eRace2
- DIS/JRU

3 Functional requirements

3.1 Train Protection

3.1.1 Driver´s Safety Device (DSD)

The vigilance device is integrated into the SCMT and therefore the DVD controls (pushbuttons on the traction lever and pedal) are managed by the SCMT control unit.

3.1.1.1 Driver´s Vigilance Device (DVD)

3.1.1.1.1 Function Design

The vigilance device monitor the presence and vigilance of the driver through the activation of the controls in the cabin.
In case the driver vigilance or presence is not confirmed the DVD switch on an optical warning, then acoustical warning and in the end it applies the emergency brake. The event is then recorded by the train recorder and a message is sent to the ground by the cab radio. The SCMT has two digital input to monitor the driver vigilance, the “primary chain” and the “secondary chain”.
Safety DVD control devices (pedal and button on traction lever) are connected to the primary chain. The secondary chain is connected to a TCMS output: the TCMS will monitor driver actions on several control (e.g IDU acknowledged, IR command, horn command ecc…) and then will inform the SCMT of the driver vigilance. Refer to RD [5] and to the following picture.

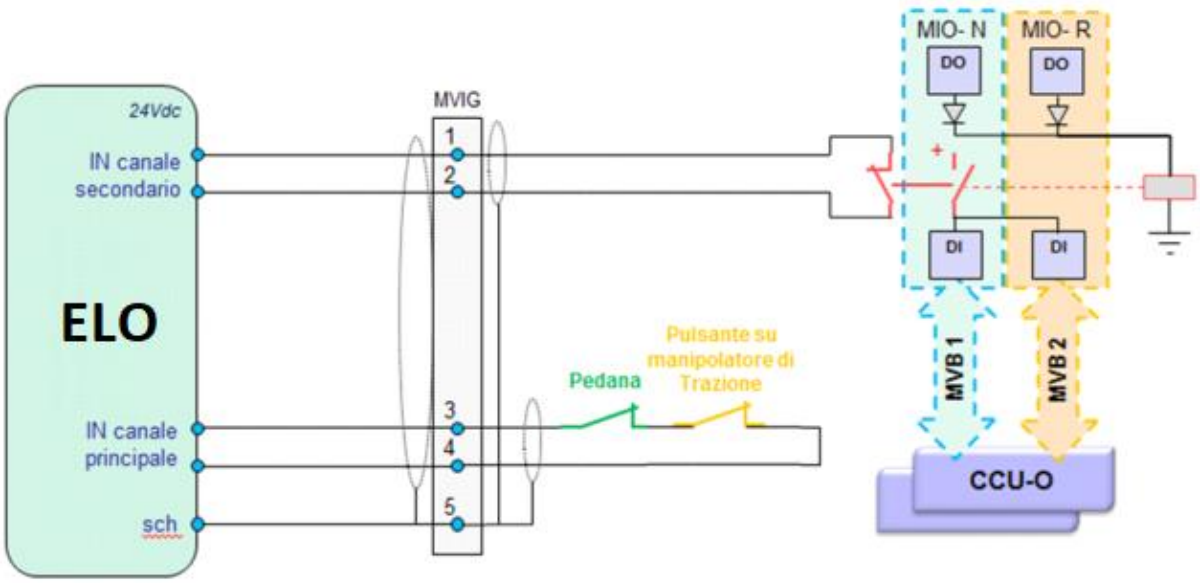


Figure 2: Vigilance device scheme

3.1.1.1.2 Normal condition

The driver confirm its presence and vigilance through the pedal or the pushbutton on the traction lever: there is no buzzer and no brake application.
The driver does not confirm its presence or vigilance: the DVD activates the buzzer and there is still no action from the driver it applies the emergency brake.

3.1.1.1.3 Failure cases

A. DVD applies the brake when not needed => The train is stopped.
B. DVD fails to apply the brake when commanded=>the train is not stopped when needed.

3.1.1.1.4 Failure detection

A. Failure self-evident.
B. Failure can be detected during the daily test initiated by the driver.

3.1.1.1.5 Failure handling

A. The DVD shall be isolated and the system shall be repaired during maintenance.
B. If a DVD failure is detected during the daily test the mission is cancelled.

3.1.1.1.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.105
Description	The driving personnel shall confirm presence and vigilance by activating the safety pedal or pushbuttons.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.638	The status of the safety pedal (operated/not operated) shall be provided to the DVD (SCMT CU) on the primary chain.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.639	The status of the control on the traction brake lever (operated/not operated) shall be provided to the DVD (SCMT CU) on the primary chain.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.640	The TCMS shall be able to monitor the vigilance of the driver and to inform the DVD (SCMT CU) through a digital output to the DVD on the “secondary chain”.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.625	When the TCMS detect driver vigilance (see concept.626) shall enerigize the relevant DO. The signal shall be to grant a LOW signal to the DVD for a time 1 s<t<1.5 s. The signal shall be then deenergized for a time 1 s<t<1.5 s. In case TCMS detect further actions of the driver a new cycle shall be provided by the TCMS to the DVD.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.626	When an action, reported in the below list, is detected by TCMS a dedicated DO shall be energized according contept.625 logic. **Movement of the traction/brake lever is detected:** The sector of 80° shall be divided in 20 sub sector and the lever movement shall be detected when the lever moves from one section to another one. A proper hysteresis shall be defined in order to avoid undue oscillation of the signal.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	Action on the automatic speed lever (ASC): The ASC moves from position N to position Vel+ or Vel-. **Action on the voltage selection selector:** The selector is moved from one position to another. **Action on LCB lever:** The lever moves from neutral position to “OPEN” or “CLOSE”. **Horn activation:** The horn command lever moves from “OFF” position to “HIGH” or to “LOW”. **Action on IDU display** Any driver action on the display such as page change request, or acknowlede of diagnostic events and alarms **Action on external light levers** Movement of levers “Full Beam" (comando luci di profondità) or “Head Lamp" (comando luci di posizione) or “Central Beam" (comando faro centrale) **Action on sanding devices push button**				
2F_09.02.Zefiro-Europe.CONCEPT.628	A feedback of the TCMS DO “driver vigilance” shall be provided to the TCMS itself for diagnostic purpose.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.627	A diagnostic event shall be generated by the TCMS in case of incongruence between DO “driver vigilance” and its feedback.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.145
Description	Emergency brake shall be applied if the driver does not confirm the presence after first an optical warning and after that a acoustic warning.
Safety level	SIL 2

The MMI is integrated in the ETCS/SCMT DMI and sound generator.

The emergency brake application is performed through the same interface as for the ATP (DVD is integrated in the ATP): the DVD will command the dedicated pneumatic plate that will cause the brake pipe discharge. Refer to §3.1.2.5.

ID	2F_09.02.Zefiro-Europe.TRS.92
Description	The DVD shall be interfaced with the JRU in order to provide data to be recorded
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.643	The DVD shall provide through MVB to the DIS/JRU the DVD EB intervention for recording purpose.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.147
Description	The DVD shall be interfaced with the Cab radio through MVB for the generation of the GSM-R alarm in case of DVD intervention

Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.96	The DVD shall provide through MVB to the cab radio the DVD EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.378	The DVD shall provide through digital output to the cab radio the DVD EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.202	The Cab radio shall send the alarm message when receive the AVD EB intervention	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.379	The status of the DVD EB intervention shall be provided to the cab radio.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.148
Description	The DVD shall get digital input to detect the driver desk status
Safety level	SIL 2

The DVD is integrated with ATP (same ATP cab status input are used, refer to §3.1.2.4).

3.1.1.2 Driver´s Vigilance Device Isolation

3.1.1.2.1 Function design

The DVD is integrated into the ATP system. Therefore the isolation of the DVD (in case of failure) equals to isolation of the ATP (ETCS/SCMT).
In the current implementation it is however possible to exclude the vigilance function through a dedicated selector to be foreseen in the area near the driver cab.

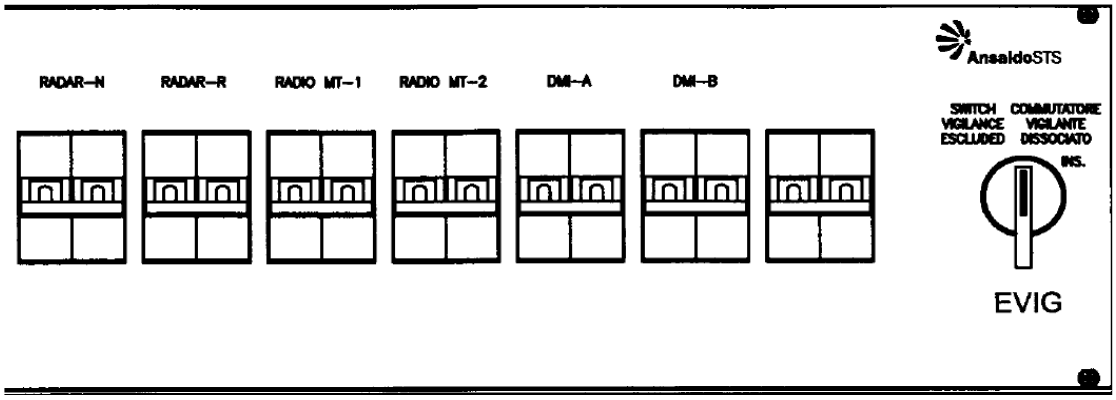


Figure 3: EVIG selector for vigilance exclusion

When the selector is in the excluded position the vigilance function is excluded and only the Italian specific function “protezione atto partenza” is performed.
When only the function “protezione atto partenza” is active the DVD supervision is limited to the moment when the train start to move, i.e the driver as to acknowledge thorugh the DVD dedicated controls that the train starts to move.
The status of the EVIG is also acquired and recorded by the DIS/JRU system.

3.1.1.2.1.1 Normal condition

In case of failure of the DVD the driver isolates it from the brakes.

3.1.1.2.1.2 Failure cases

A. DVD is not isolated from the brake when the isolation switch is in isolation position.
B. DVD is isolated from the brake when the isolation switch is in normal position.

3.1.1.2.1.3 Failure detection

A. Failure is self-evident. The train remains braked.
B. Failure can be detected during the daily test initiated by the driver. This failure however can be expected only after switch movement from isolation to normal position, therefore after the action the execution of the test is expected.

3.1.1.2.1.4 Failure handling

A. To release brake application pneumatic valve could be bypassed.
B. The failure shall be repaired during maintenance.

3.1.1.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.111
Description	It shall be possible to Bypass the EB applied by the DVD in case of a faulty DVD system.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.381	A switch named EVIG on the cab shall be foreseen to exclude the vigilance function	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.382	The status of the switch EVIG shall be provided to the DVD (SCMT CU).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.648	The status of the switch EVIG shall be provided to the DIS/JRU for recording purpose	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.254
Description	The DVD test shall be started manually by the driver in the active cab.
Safety level	SIL 0

3.1.2 Automatic Train Protection System (ATP)

3.1.2.1 Visualization

3.1.2.1.1 Function design

3.1.2.1.1.1 Normal condition

The ETCS DMI shall visualize the time received by the DIS/JRU system (GPS time).
The speed is normally shown on the ETCS DMI.
In case of ATP system failure and consequent isolation the speed will be displayed on the IDU.

3.1.2.1.1.2 Failure cases

A. ATP system wrongly detected as isolated by the TCMS => speed is displayed contemporaneously on ATP DMI and on IDU.
B. ATP system wrongly detected as not isolated by the TCMS => speed signal is not available to the driver.

3.1.2.1.1.3 Failure detection

A. Failure self-evident.
B. Failure self-evident.

3.1.2.1.1.4 Failure handling

A. Failure isolated and repaired during maintenance.
B. Failure isolated and repaired during maintenance.

3.1.2.1.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.154
Description	The train speed shall be displayed on the diagnostic display in case of absence or degradation of the ATP system.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.317	If the ATP is isolated by the isolation switch (CEA) the train speed shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.734	The TCMS shall acquire the status of the ATP isolation switch (CEA). Refer to RD [1].	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.385	The CEA status shall be recorded by the DIS/JRU	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.386	The status of the CEA shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.174
Description	The ATP system shall provide the time to the train for syncronization to all equipment onboard.
Safety level	SIL 0

The time for on-board synchronization will come from DIS/JRU and be sent to TCMS. TCMS will use the MCG via SNTP to do the synchronization.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.651	The DIS/JRU shall send the time in UTF format to TCMS for time synchronization on-board.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.652	The TCMS shall use the time from the DIS/JRU for on-board time synchronization	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.2 ATP Direction control

3.1.2.2.1 Function design

The ATP monitor the direction selected by the driver and the actual train movement: in case of incoherence it applies the emergency brake.

3.1.2.2.1.1 Normal condition

The selected direction is neutral (no selection), the train moves forward or backward => ATP emergency brake application.
The selected direction is forward, the train moves backward => ATP emergency brake application.
The selected direction is backward, the train moves forward => ATP emergency brake application.

3.1.2.2.1.2 Failure cases

A. Loss of ATP emergency brake application.

3.1.2.2.1.3 Failure detection

A. Failure detected during system power on self-test.

3.1.2.2.1.4 Failure handling

A. Failure isolated and repaired during maintenance.

3.1.2.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.158
Description	The requirements of signals necessary for ATP to control the driving direction of the train during ATP system supervision shall be defined. The defined driving directions could be: - Forward Direction - Backward Direction
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.653	The TCMS shall provide to the ETCS the signal position Forward of the direction controller through MVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.654	The TCMS shall provide to the ETCS the signal position Backward of the direction controller through MVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.655	If the direction controller is set to Neutral then Forward and Backward direction signals over the MVB shall be reset to the ATP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.3 ATP Isolation

3.1.2.3.1 Function design

In case of ATP failure, the ATP itself applies the emergency brake (fail safe) therefore in this case there shall be the possibility to isolate the device to be able to move the train: the ATP can be isolated through a sealed switch.

3.1.2.3.1.1 Normal condition

In case of failure of the ATP the driver isolates it from the brakes.

3.1.2.3.1.2 Failure cases

A. ATP is not isolated from the brake when the isolation switch is in isolation position.
B. ATP is isolated from the brake when the isolation switch is in normal position.

3.1.2.3.1.3 Failure detection

A. Failure is self-evident. The train remains braked.
B. Failure can be detected during power on test. This failure however can be expected only after switch movement from isolation to normal position, therefore after the action the execution of the test is expected.

3.1.2.3.1.4 Failure handling

A. To release brake application pneumatic valve could be bypassed.
B. The failure shall be repaired during maintenance.

3.1.2.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.120
Description	ATP isolation shall be possible using a sealed isolation switch.
Safety level	N/A

3.1.2.4 ATP Start up Mission during Cab activation

3.1.2.4.1 Function design

3.1.2.4.1.1 Normal condition

The system shall propose to the driver the start of mission when the cab is activated.

3.1.2.4.1.2 Failure cases

A. ATP does not detect the cab activation=> the mission shall be cancelled.

3.1.2.4.1.3 Failure detection

A. Failure is self-evident.

3.1.2.4.1.4 Failure handling

A. Failure isolated and repaired during maintenance.

3.1.2.4.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.124
Description	ATP shall automatically start if cab has been activated. If cab has been activated ATP in the activated cab shall get a signal which allows the ATP to start its Mission.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.656	The status “Cab Active shall be provided to the ATP through 1 switch. Refer to RD [1]	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.657	The status “cab not Active” shall be provided to the ATP through 1 switch. Refer to RD [1]	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

3.1.2.5 ATP Emergency Brake control

3.1.2.5.1 Function design

3.1.2.5.1.1 Normal condition

The ATP applies the emergency brake commanding directly the dedicated local electro valves: the brake pipe is then discharged from the active cab.

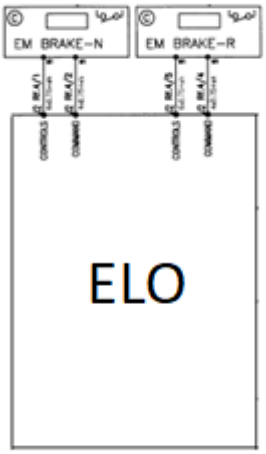


Figure 4: ELO - Pneumatic plates interface

In parallel to the pneumatic plate command the ATP commands also the inhibition of the refilling of the brake pipe: this command will be used to de energize the emergency brake loop.

3.1.2.5.1.2 Failure cases

A. Loss to de energize the emergency brake loop => the brake pipe will be discharged only from the active cab therefore the equivalent time of the brake application will be greater.

3.1.2.5.1.3 Failure detection

A. Failure detected by the TCMS.

3.1.2.5.1.4 Failure handling

A. Failure isolated and repaired during maintenance.

3.1.2.5.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.130
Description	ATP shall be integrated in the train's electrical and pneumatical circuits in such a manner that it can order Emergency Brake.
Safety level	SIL 4

ID	2F_09.02.Zefiro-Europe.TRS.170

Description	When ordered by ATP Emergency brake shall be applied in the whole train by cutting the power supply from the magnet valve (reverse) of the pneumatic brake.
Safety level	SIL 4

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.660	The ATP will order emergency brake commanding its dedicated pneumatic plate	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.661	In parallel to the command to the Pneumatic plate the ATP will command the Brake pipe refilling inhibition through a digital output. Refer to RD [1]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.662	When the ATP command the brake pipe refilling inhibition Emergency brake shall be applied in the whole train by de-energising the Emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.329	If ATP orders Emergency brake an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.6 ATP Service Brake control

3.1.2.6.1 Function design

3.1.2.6.1.1 Normal condition

The ATP will command the service brake when needed (First Line of Intervention), the TCMS will forward the request to the brake system that shall apply the service brake.
Note: in case the service brake command will not be applied or in case the speed reduction is not enough to avoid crossing the second line of intervention the ATP will command the emergency brake.

3.1.2.6.1.2 Failure cases

A. Service brake not applied when commanded by the ATP.

3.1.2.6.1.3 Failure detection

A. Failure self-evident: EB applied by the ATP.

3.1.2.6.1.4 Failure handling

A. Failure isolated and repaired during maintenance.

3.1.2.6.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.132
Description	ATP shall be able to order Service Brake.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.663	The ATP shall command the full service brake through MVB.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.664	The TCMS shall apply service brake when ordered by ATP according to Brake blending	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_09.02.Zefiro-Europe.CONCEPT.387	The Italian national ATP (SCMT) shall command the service brake through a digital output (free voltage contact)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.388	The TCMS shall acquire the service brake command	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.389	The TCMS shall apply service brake when ordered by ATP (SCMT)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.7 Traction Cut-off

3.1.2.7.1 Function design

3.1.2.7.1.1 Normal condition

The ATP will command the SW traction block when needed (phase and voltage transition, speed limitation), the TCMS will forward the request to the traction system that shall remove the traction effort.
Also the DIS/JRU system shall be able to block the traction in specific circumstances (for example in case the speed is over the admitted threshold and the ATP is isolated, i.e the switch CEA has been operated).

3.1.2.7.1.2 Failure cases

A. Traction effort wrongly not removed => the MCB will be opened with traction effort.

3.1.2.7.1.3 Failure detection

A. Failure not detected.

3.1.2.7.1.4 Failure handling

A. Failure isolated and repaired during maintenance.

3.1.2.7.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.136
Description	If ATP system orders brake (Emergency Brake / Service Brake) ATP will send a signal which shall lead to software block of traction demand in the whole train.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.666	The ATP shall be able to command SW traction block when necessary through a digital output. Refer to RD [1]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.667	The TCMS shall acquire the SW traction block command	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.121	If ATP SW traction block command is applied AND the switch CEA is not operated then Traction block shall be applied in the whole train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.668	The DIS/JRU shall be able to command SW traction block (trough a digital output)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.669	The TCMS shall acquire the SW traction block command from the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.670	A sealed switch named CETD on the cab shall be foreseen to bypass the DIS/JRU SW traction block	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 0

2F_09.02.Zefiro-Europe.CONCEPT.671	The DIS/JRU shall acquire the status of the switch CETD for recording purpose	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.396	The TCMS shall acquire the status of the switch CETD	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.672	When the switch CETD is operated the lamp “avaria DIS/JRU” shall be switched on (refer to 2F_09.02.Zefiro-Europe.CONCEPT.711)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.673	If DIS/JRU SW traction block command is applied by the DIS/JRU related to the active cab AND the switch CETD is not operated then Traction block shall be applied in the whole train by the TCMS. Note: traction block command of DIS/JRU of not active cabs shall ne ignored by the TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.8 Main Circuit breaker command

3.1.2.8.1 Function design

3.1.2.8.1.1 Normal condition

The ATP will command the main circuit breaker command when needed (phase and voltage transition, speed limitation); the TCMS will forward the request to the traction system that will command the MCB opening.
The ATP will command the MCB opening through two different command for redundancy (MVB and Digital output) to guarantee the command execution. It is then a “Train” task to assure the execution of the command: the digital output can be acquired by the TCMS and the command managed by the TCMS SW or it can be directly used to drive a Trail line to command the MCB opening via HW.

3.1.2.8.1.2 Failure cases

3.1.2.8.1.3 Failure detection

3.1.2.8.1.4 Failure handling

3.1.2.8.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.256
Description	The ATP shall give additional information to open the main circuit breaker. On leaving the separation section the ATP shall cause main circuit breaker to be closed.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.674	The ATP shall be able to command main circuit breakers opening when necessary thorugh a digital output. Refer to RD [1]	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.437	The TCMS shall acquire the MCB opening command	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.436	The TCMS shall command the MCB opening when ordered by ATP (SCMT) via digital output	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.435	In redundancy to the digital command the ATP shall be able to command main circuit breakers opening thorugh the MVB	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.	The TCMS shall command the MCB opening when ordered by ATP (SCMT) via MVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

434					

3.1.2.9 Phase Transition

3.1.2.9.1 Function design

3.1.2.9.1.1 Normal condition

The phase transition in FS is completely automatic while in SR and when the ATP is isolated it shall be realized manually.
The ATP will command to the TCMS the SW traction block and MCB opening in advance to the neutral section and will remove these commands at the end of the neutral section.
In order to reduce the time with the SW traction block and open MCB the TCMS shall also communicate to the ATP which is the raised 25kV pantograph (front or rear pantograph): the ATP will then command the SW traction block and MCB opening before the active pantograph reaches the neutral section and will give the command MCB closure and traction reestablishment when the active pantograph has left the neutral section.

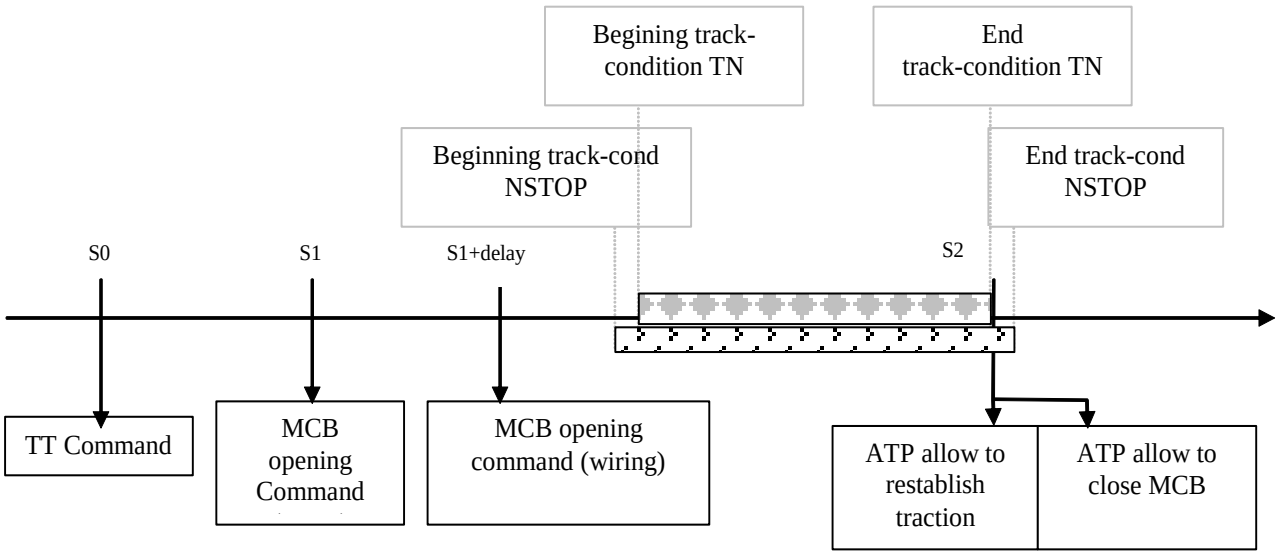


Figure 5: Phase transition sequence

3.1.2.9.1.2 Failure cases

3.1.2.9.1.3 Failure detection

3.1.2.9.1.4 Failure handling

3.1.2.9.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.258
Description	The phase separation sections and separation section sequences shall be activated by External sensors e.g. ATP or track side magnets.

Safety level	SIL 0

The ATP will compute the location where to cut off the traction and will command it as described in §3.1.2.7
The ATP will compute the location where to open the MCB and will command it as described in §3.1.2.8
The computation of the location to command and revoke the SW traction block and MCB opening command is based on the fact that the ETCS know which is the active pantograph (front or back pantograph).

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.451	The TCMS shall inform the ATP through MVB which is the active pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

When the condition for the traction restoration are met the ATP will remove the SW traction block and MCB opening command.

3.1.2.10 Voltage Transition

3.1.2.10.1 Function design

3.1.2.10.1.1 Normal condition

The voltage transition in FS is automatic and it requires to the driver only to select the new voltage through the line voltage selector on the driver desk at the end of the neutral section.
In SR and when the ATP is isolated it shall be realized manually.
The ATP will command to the TCMS the SW traction block, MCB opening and pantograph lowering in advance to the neutral section and will remove these commands at the end of the neutral section.

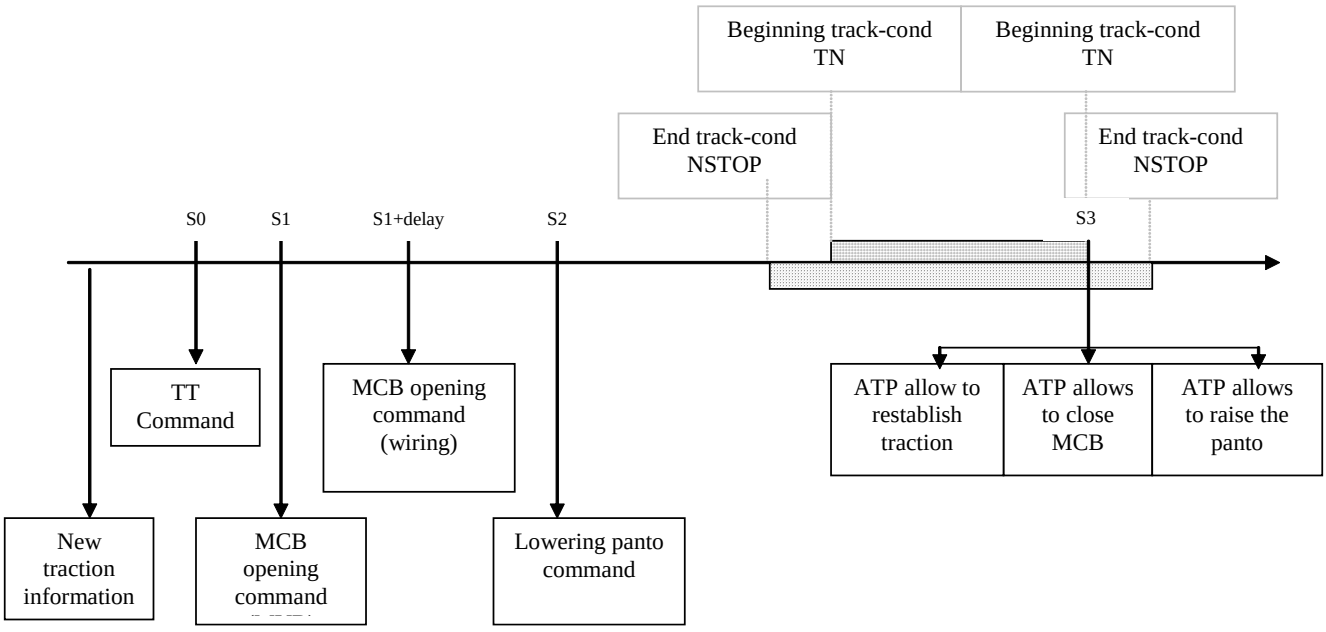


Figure 6: Voltage transition sequence

3.1.2.10.1.2 Failure cases

3.1.2.10.1.3 Failure detection

3.1.2.10.1.4 Failure handling

3.1.2.10.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.260
Description	Where phase separation sections require pantographs to be lowered and subsequently raised, these additional actions are permitted to be automatically intiated by ATP. These functions shall respond to input signals from the control-command and signalling system.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.457	The ATP shall be able to send the new voltage value to the TCMS via MVB (change power command)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0

The ATP will compute the location where to cut off the traction.
The ATP will compute the location where to open the MCB and will command it as described in §3.1.2.8.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.576	The ATP will compute the location where to lower the pantograph and will command it through the MVB (Raising pantograph command)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.460	The TCMS shall lower the pantograph when commanded by the ATP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

When the condition for the traction restoration are met the ATP will remove the SW traction block, MCB opening and lower the pantograph command

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.463	If the ATP removes the SW traction block, MCB opening and lower the pantograph command the TCMS shall raise the pantograph if the driver has correctly selected the voltage type and if all safety condition of its raising are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.464	If the ATP removes the SW traction block, MCB opening and lower the pantograph command and the pantograph has been raised the TCMS shall close the MCB and restore the traction only if all safety condition are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.465	If the line voltage system selected by the driver does not correspond to the line voltage system reported by the ATP an event shall be set and the pantograph shall not be allowed to be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.11 Sleeping

Sleeping functionality is not implemented at this moment for V300Zefiro Italy project. For this functionality the ATP shall receive 2 antivalent-contacts for future uses.

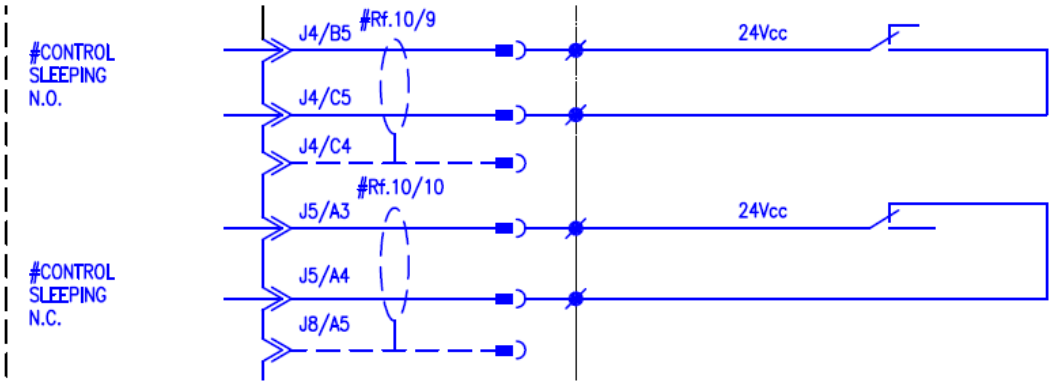


Figure 7: Sleeping commands inputs

NB: when the function will be implemented, some changes at train level will be required in order to manage properly this information.

3.1.2.11.1 Function design

3.1.2.11.1.1 Normal condition

The meaning of the contacts is the following:

NO contact	NC contact	Sleeping command
Open	Closed	Sleeping not Requested
Closed	Open	Sleeping Requested
Open	Open	Error
Closed	Closed	Error

3.1.2.11.1.2 Failure cases

3.1.2.11.1.3 Failure detection

3.1.2.11.1.4 Failure handling

3.1.2.11.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.343
Description	ATP shall be able to manage Sleeping Information (predisposition).
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.690	The sleeping command shall be provided to the ATP with the meaning: SLEEPING_COMMAND_NO: OPEN: Sleeping not Requested CLOSED: Sleeping Requested	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 1

2F_09.02.Zefiro-Europe.CONCEPT.691	The sleeping command shall be provided to the ATP with the meaning: SLEEPING_COMMAND_NC: OPEN: Sleeping Requested CLOSED: Sleeping not Requested	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 1

3.1.2.12 Train Integrity

Train integrity functionality is not implemented at this moment for V300Zefiro Italy project. For this functionality the ATP shall receive 2 antivalent-contacts for future uses.

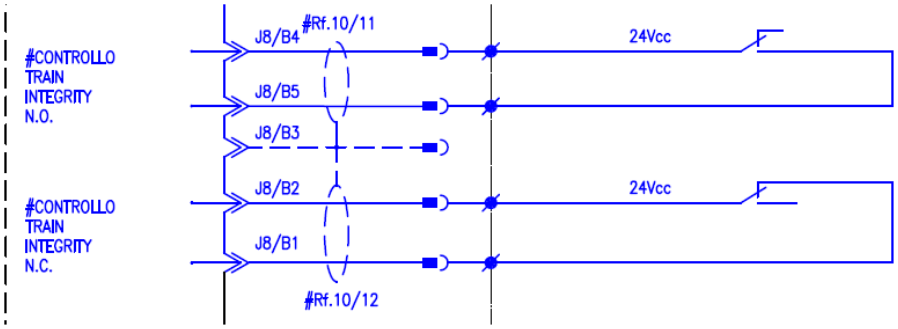


Figure 8: Train Integrity commands inputs

NB: when the function will be implemented, some changes at train level will be required in order to manage properly this information.

3.1.2.12.1 Function design

3.1.2.12.1.1 Normal condition

The meaning of the contacts is the following:

NO contact	NC contact	Train integrity input
Open	Closed	Train Integrity NOT OK
Closed	Open	Train Integrity OK
Open	Open	Error/ Train Integrity NOT OK
Closed	Closed	Error/ Train Integrity NOT OK

3.1.2.12.1.2 Failure cases

3.1.2.12.1.3 Failure detection

3.1.2.12.1.4 Failure handling

3.1.2.12.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.345
Description	ATP shall be able to manage Train Integrity Information (predisposition).
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.705	The Train Integrity Status shall be provided to the ATP with the meaning: TRAIN_INTEGRITY_NO: OPEN: TRAIN INTEGRITY NOT OK CLOSED: TRAIN INTEGRITY OK	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.706	The Train Integrity Status shall be provided to the ATP with the meaning: TRAIN_INTEGRITY _NC: OPEN: TRAIN INTEGRITY OK CLOSED: TRAIN INTEGRITY NOT OK	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		SIL 0

3.1.2.13 Train Configuration

3.1.2.13.1 Function design

Train configuration takes in account the performance of the vehicle in case of nominal or degraded condition in terms of isolated bogie → percentage of braking mass/maximum speed.
Possible configurations take in account from nominal condition (all bogies available) up to 4 isolated bogies.

3.1.2.13.1.1 Normal condition

No bogie isolated.

3.1.2.13.1.2 Failure cases

From 1 to 4 isolated bogies.

3.1.2.13.1.3 Failure detection

N/A

3.1.2.13.1.4 Failure handling

Isolation of the faulty bogie/bogies and selection of the appropriated train data set.

3.1.2.13.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.754	The train data entry shall be “Swhitchable type” data entry.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.755	The ETCS DMI shall show the button to switch from fixed to flexible data Entry with the text: “Enter Data”	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.756	The ETCS DMI shall show the button to switch from flexible to fixed data Entry with the text: “Select Type”	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_09.02.Zefiro-Europe.CONCEPT.757</td><td>The ATP shall foresee at least the following dataset, in order to allow the driver to select the correct configuration during data entry. in base of the effective braking capacity of the train:
- Normal condition (no isolated bogie)
- 1 isolated bogie
- 2 isolated bogies
- 3 isolated bogies
- 4 isolated bogies

All parameters associated to the configuration above are reported in a separated document,</td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 4</td></tr>
<tr><td>2F_09.02.Zefiro-Europe.CONCEPT.758</td><td>The driver is allowed to modify the following data if flexible data entry is selected:
<table><tr><th>Input Field</th><th>Keyboard</th></tr><tr><td>Train category</td><td>Dedicated keyboard</td></tr><tr><td>Length (m)</td><td>Numeric</td></tr><tr><td>Brake percentage</td><td>Numeric</td></tr><tr><td>Maximum speed (km/h)</td><td>Numeric</td></tr><tr><td>Axle load category</td><td>Dedicated keyboard</td></tr><tr><td>Airtight</td><td>Dedicated keyboard
Touch screen technology:
Key ‘7’ = No
Key ‘8’ = Yes</td></tr><tr><td>Loading gauge</td><td>Dedicated keyboard</td></tr></table></td><td>Derived Requirement</td><td>4S_09.06.-.Automatic Train Protection Systems</td><td></td><td>SIL 4</td></tr>
</table>

3.1.3 Juridical Recorder Unit (JRU)

3.1.3.1 Function design

3.1.3.1.1 Normal condition

The train recorder shall record all signal coming from the train through MVB (TCMS) or through digital input.

3.1.3.1.2 Failure cases

3.1.3.1.3 Failure detection

3.1.3.1.4 Failure handling

3.1.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.169
Description	Signals to be recorded as a minimum: · "Clear to proceed" from the ATP system · "Restrictive signal aspect" from the ATP system · "Acknowledgement" from the ATP system · Time · Speed · Brake applied · EB applied · ATP cab switch · The interaction between the driver and the ATP system in terms of actions done by the personnel and of

	communications sent to the driver by means of visual and/or acoustic signals · The data which the onboard technological system exchanges with the ground sub-systems with which it communicates · Distance · The data that the ATP exchanges with the equipment and systems of the train · The data pertaining to the ATP functional status.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.707	The TCMS shall provide to the DIS/JRU data to be recorded though MVB according to RD [3] and RD [4].	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.708	The DIS/JRU shall record data received through MVB according to RD [3].	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.709	The status of digital input shall be provided to the DIS/JRU according to RD [3].	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.710	The DIS/JRU shall record digital input status according to RD [3].	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.175
Description	A faulty JRU shall be reported to the driver.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.711	The DIS/JRU shall command a lamp “avaria DIS/JRU” on the driver desk in case of failure	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.712	The lamp “avaria DIS/JRU” shall be switched on also in case of DIS/JRU circuit breaker is tripped	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.713	An event shall be generated if the DIS/JRU is faulty	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.714	The DIS/JRU shall command a lamp on the driver desk in case of recorder memory is full (70% of memory)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.336	An event shall be generated in case of recorder memory is full	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.715	The DIS/JRU shall be switched on as soon as the battery are connected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.623	The status of the bus 180 m shall be provided to the DIS/JRU for the management of the sleeping function.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.469	The DIS/JRU shall acquire the status of the cab (active/not active)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.468	The status of the cab (activated/not activated) shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

3.1.4 Cab Radio

3.1.4.1 Function design

3.1.4.1.1 Normal condition

The cab is equipped with a GSM-R radio for train ground voice communication. The radio will be connected to the intercom and PA system and will have proper interface to the vigilance device for the alarm generation in case of DVD intervention.
The radio shall be powered as soon as the battery contactors are closed but it shall be possible to power it also when the train is off through a dedicated switch.
The failure of the radio will be clearly indicated on the driver desk.

3.1.4.1.2 Failure Cases

3.1.4.1.3 Failure Detection

3.1.4.1.4 Failure Handling

3.1.4.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.262
Description	The cab radio shall be compliant to RFI standard RFI DTC CSI SR OR 10 003 A, “Sistema Tecnologico di Bordo - Apparato "Cab Radio": requisiti nazionali ad integrazione dei requisiti europei”
Safety level	N/A

Power management

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.483	The cab radio shall be switched on as soon as the battery are connected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.484	There shall be a switch in the cabin in order to power on the cab radio when the battery are disconnected	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

Alarm lamp

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.735	In case the cab radio is not powered or in case of cab radio major fault the cab radio shall provide this information trough a contact free of voltage (digital output Cab radio not available, alarm light mng, refer to RD [2])	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.487	When the cab radio is not available a light indication shall be switched on the driver desk	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.488	The alarm lamp status shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

Alarm message

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.	The cab radio shall acquire the EB status sent from the DVD (refer to §3.1.1 and to to RD [2],	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0

736	input DSD alarm input).				

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Input (ATP/DVD)

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.716	SafetyPedal	L	2F_09.02.Zefiro-Europe.CONCEPT.638
2F_09.02.Zefiro-Europe.CONCEPT.717	ManipulatorControl	L	2F_09.02.Zefiro-Europe.CONCEPT.639
2F_09.02.Zefiro-Europe.CONCEPT.538	EVIG	L	2F_09.02.Zefiro-Europe.CONCEPT.382
2F_09.02.Zefiro-Europe.CONCEPT.718	CabActive	L	2F_09.02.Zefiro-Europe.CONCEPT.656
2F_09.02.Zefiro-Europe.CONCEPT.719	CabNotActive	L	2F_09.02.Zefiro-Europe.CONCEPT.657

Inputs (DIS/JRU)

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.720	EVIG	L	2F_09.02.Zefiro-Europe.CONCEPT.648
2F_09.02.Zefiro-Europe.CONCEPT.721	CETD	L	2F_09.02.Zefiro-Europe.CONCEPT.671
2F_09.02.Zefiro-Europe.CONCEPT.539	CEA	L	2F_09.02.Zefiro-Europe.CONCEPT.386
2F_09.02.Zefiro-Europe.CONCEPT.540	Cab Status	L	2F_09.02.Zefiro-Europe.CONCEPT.468
2F_09.02.Zefiro-Europe.CONCEPT.541	Pneumatic plate 1 status	L	
2F_09.02.Zefiro-Europe.CONCEPT.542	Pneumatic plate 2 status	L	

Input (Cab radio)

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.546	DVD intervention	L	2F_09.02.Zefiro-Europe.CONCEPT.379

Outputs (ATP)

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.553	DVD intervention	L	2F_09.02.Zefiro-Europe.CONCEPT.378
2F_09.02.Zefiro-Europe.CONCEPT.722	Brake pipe refilling inhibition	L	2F_09.02.Zefiro-Europe.CONCEPT.661
2F_09.02.Zefiro-Europe.CONCEPT.558	Service brake command	L	2F_09.02.Zefiro-Europe.CONCEPT.387
2F_09.02.Zefiro-Europe.CONCEPT.723	SW traction block	L	2F_09.02.Zefiro-Europe.CONCEPT.666
2F_09.02.Zefiro-Europe.CONCEPT.724	Main Circuit breaker command	L	2F_09.02.Zefiro-Europe.CONCEPT.674
2F_09.02.Zefiro-Europe.CONCEPT.555	Cab radio not available	L	

Output (DIS/JRU)

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.725	SW traction block DIS/JRU	L	2F_09.02.Zefiro-Europe.CONCEPT.668
2F_09.02.Zefiro-Europe.CONCEPT.726	Avaria DIS/JRU	L	2F_09.02.Zefiro-Europe.CONCEPT.711
2F_09.02.Zefiro-Europe.CONCEPT.727	70%memory full	L	2F_09.02.Zefiro-Europe.CONCEPT.714

Outputs (Cab Radio)

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.737	Cab radio major fault	L	2F_09.02.Zefiro-Europe.CONCEPT.735

4.1.2 TCMS SW

Inputs

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.738	CEA Status	L	2F_09.02.Zefiro-Europe.CONCEPT.734
2F_09.02.Zefiro-Europe.CONCEPT.569	Service brake	L	2F_09.02.Zefiro-Europe.CONCEPT.388
2F_09.02.Zefiro-Europe.CONCEPT.728	SW traction block ATP	L	2F_09.02.Zefiro-Europe.CONCEPT.667
2F_09.02.Zefiro-Europe.CONCEPT.729	SW traction block DIS/JRU	L	2F_09.02.Zefiro-Europe.CONCEPT.669
2F_09.02.Zefiro-Europe.CONCEPT.571	CETD		2F_09.02.Zefiro-Europe.CONCEPT.396

2F_09.02.Zefiro-Europe.CONCEPT.629	Driver vigilance feedback		2F_09.02.Zefiro-Europe.CONCEPT.627

Outputs

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.31	Driver Vigilance Command		2F_09.02.Zefiro-Europe.CONCEPT.625

4.2 Bus Interface

ATP to TCMS SW

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.730	Service brake command (SSBcomfreser)	L	2F_09.02.Zefiro-Europe.CONCEPT.663
2F_09.02.Zefiro-Europe.CONCEPT.574	Main Switch command driving (SSBpillR)	L	2F_09.02.Zefiro-Europe.CONCEPT.435
2F_09.02.Zefiro-Europe.CONCEPT.575	Raising pantograph command (SSBcompanto)	L	2F_09.02.Zefiro-Europe.CONCEPT.576
2F_09.02.Zefiro-Europe.CONCEPT.577	Change power command (SSBcambcaten)	A	2F_09.02.Zefiro-Europe.CONCEPT.457

TCMS SW to ATP

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.731	Forward (flglevainvert)	A	2F_09.02.Zefiro-Europe.CONCEPT.653
2F_09.02.Zefiro-Europe.CONCEPT.732	Backward (flglevainvert)	A	2F_09.02.Zefiro-Europe.CONCEPT.654
2F_09.02.Zefiro-Europe.CONCEPT.578	Pantograph status(flgTrolley)	A	2F_09.02.Zefiro-Europe.CONCEPT.451

TCMS SW DIS/JRU
According to RD [3]

5 References

RD	Code	Title	Source	Revision
[1]	B60A.0100807.L03.00IT	ERTMS system BL3 – Wiring layout and instructions for specific application on the V300 Zefiro	HSTS	Rev 00.00
[2]	S20002.03.1005TR	Cab radio functional diagram	Leonardo	01.00

[3]	FP02P011037B	SISTEMA INFORMATIVO DI CONDOTTA: ELENCO METADATI	HR	latest
[4]	EC13P011957B	DIS/JRU - DESCRIPTION OF DATA RECEIVED FROM THE TRAIN	HR	latest
[5]	EC01P014399B	Sistema vigilante - Definizione segnale di reiterazione gestito da TCMS	HR	Rev 00